

Online Resource 4-Major, minor and trace element concentrations in volcanic rocks

Table 1: XRF data for volcanic rocks.

	SiO2	Al2O3	Fe2O3(t)	FeO	MnO	MgO	CaO	Na2O	K2O	TiO2	P2O5	SO3	Cr2O3	NiO	LOI	Sum			
	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)	(wt%)			
16B10	58.6	18.2	6.09	4.73	0.01	1.95	4.46	2.00	2.32	0.80	0.08	0.13	0.01	0.01	5.07	99.7			
16B11	50.3	19.8	9.05	7.03	0.03	4.92	3.57	4.32	1.27	1.31	0.33	0.05	0.03	0.01	4.77	99.8			
16B13	46.5	15.5	8.48	6.59	0.14	6.00	13.6	0.11	0.01	0.88	0.17	0.06	0.05	0.02	8.40	100			
	Ba	Co	Cr	Cs	Cu	Ga	Nb	Ni	Pb	Rb	Sc	Sr	Th	U	V	W	Y	Zn	Zr
	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)	(µg/g)
16B10	2160	29.5	18.7	1.80	43.8	21.4	28.1	58.2	16.7	55.5	22.7	229	20.5	2.40	142	33.8	48.2	93.4	352
16B11	559	20.4	174	2.30	36.4	21.0	32.5	60.5	12.2	31.7	33.9	72.2	3.62	2.35	233	13.5	41.8	147	87.3
16B13	127	37.9	400	bdl	36.2	14.2	21.0	129	4.40	0.40	26.9	67.7	2.47	2.15	230	12.1	18.9	58.4	87.7

Table 2: Trace element data (µg/g) for volcanic rock samples gained by LA-ICPMS analysis on whole-rock glass beads obtained by Ir-strip heater procedure. Further given are element concentrations for the QCMs together with their respective reference values (GeoReM preferred values).

Sample	Ti	V	Cr	Ni	Rb	Sr	Nb	Ba	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Ta	Th	U
16B10 (n=6)	3550	104	12.2	39.5	49.5	172	21.2	1460	26.5	55.5	6.40	25.6	5.81	1.39	5.92	0.86	5.53	1.11	3.07	0.40	2.67	0.35	1.05	14.64	1.32
16B11 (n=5)	6380	190	116	46.8	30.9	56.0	27.1	480	25.7	36.8	4.61	18.4	4.22	1.24	5.23	0.85	5.85	1.20	3.58	0.50	3.24	0.47	1.58	2.55	2.25
16B13 (n=6)	5270	240	345	131	0.18	68.4	20.3	105	13.7	25.8	3.04	12.5	2.96	0.98	3.19	0.50	3.24	0.69	2.08	0.29	1.96	0.29	1.10	2.11	0.69
16B10 SD	161	4.04	0.89	1.76	2.20	7.44	1.00	68.0	1.36	2.85	0.34	1.44	0.38	0.10	0.42	0.06	0.36	0.07	0.21	0.03	0.23	0.03	0.07	0.84	0.09
16B11 SD	288	7.33	5.29	2.06	1.39	2.69	1.28	22.5	1.33	1.90	0.25	1.06	0.32	0.09	0.39	0.06	0.38	0.08	0.24	0.04	0.26	0.04	0.10	0.16	0.14
16B13 SD	238	9.22	14.7	5.29	0.03	3.15	0.97	5.09	0.72	1.33	0.17	0.75	0.25	0.08	0.27	0.04	0.24	0.05	0.16	0.03	0.19	0.03	0.07	0.14	0.05

		Ti	V	Cr	Ni	Rb	Sr	Nb	Ba	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Ta	Th	U
NIST SRM 612	Mean	40.0	39.9	36.9	41.0	33.1	84.4	41.4	40.3	39.6	40.3	40.5	39.4	42.3	38.5	42.8	41.0	40.5	42.6	43.4	41.4	42.9	41.5	41.0	42.8	39.0
	SD	8.26	1.16	1.05	0.91	1.03	2.20	1.10	3.38	1.18	1.24	1.29	1.52	1.02	1.24	0.81	0.76	0.88	0.75	1.07	0.86	1.38	1.10	0.89	1.13	1.48
	Ref. value	44.0	38.8	36.4	38.8	31.4	78.4	38.9	39.3	36	38.4	37.9	35.5	37.7	35.6	37.3	37.6	35.5	38.3	38	36.8	39.2	37	37.6	37.79	37.38
USGS BCR-2G	Mean	14400	447	15.9	12.6	48.8	356	12.7	712	26.3	54.8	6.88	30.6	6.99	2.14	7.39	1.06	6.83	1.35	3.90	0.52	3.67	0.51	0.75	6.42	1.72
	SD	514	14.0	0.77	0.79	2.07	11.6	0.41	23.9	1.52	2.01	0.28	2.13	0.47	0.17	0.76	0.05	0.53	0.13	0.32	0.08	0.23	0.05	0.04	0.36	0.07
	Ref. value	14100	425	17	13	47	342	12.5	683	24.7	53.3	6.7	28.9	6.59	1.97	6.71	1.02	6.44	1.27	3.7	0.51	3.39	0.503	0.78	5.9	1.69

USGS GSD-1G	Mean	7960	44.2	40.5	60.3	39.7	70.8	45.1	71.5	40.5	42.0	46.2	46.4	49.5	42.6	51.0	48.5	54.1	51.1	40.9	51.6	52.9	53.6	43.1	42.7	41.5
	SD	180	1.08	4.17	2.70	0.94	2.72	1.24	2.80	1.61	1.44	1.66	1.83	1.91	1.40	1.89	1.83	2.26	2.04	1.44	1.66	2.12	2.06	1.22	1.00	1.09
	Ref. value	7434	44	42	58	37.3	69.4	42	67	39.1	41.4	45	44.7	47.8	41	50.7	47	51.2	49	40.1	49	50.9	51.5	40	41	41
MPI-DING StHs6/80-G	Mean	4280	86.7	11.8	20.9	30.0	501	6.67	312	12.2	26.2	3.21	13.3	2.99	1.03	2.64	0.37	2.29	0.42	1.21	0.18	1.14	0.16	0.40	2.33	1.34
	SD	122	1.89	1.45	0.64	0.37	12.7	0.20	7.98	0.37	0.58	0.09	0.61	0.22	0.08	0.32	0.04	0.12	0.04	0.08	0.02	0.11	0.01	0.03	0.11	0.74
	Ref. value	4200	90.3	16.9	23.7	30.7	482	6.94	298	12	26.1	3.2	13	2.78	0.953	2.59	0.371	2.22	0.42	1.18	0.172	1.13	0.168	0.42	2.28	1.01